

An efficient and accurate adaptive time-stepping method for the Allen–Cahn equation

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ABSTRACT

We propose a time-stepping scheme to efficiently and accurately solve the Allen–Cahn (AC) equation, which is one of the fundamental partial differential equations used in modeling phase separation and interface dynamics. The proposed method dynamically changes the time step based on the maximum norm of consecutive evolutions of the solution, which ensures both computational efficiency and accuracy. A stability analysis is provided. Numerical tests confirm that the adaptive approach outperforms fixed time-stepping schemes in terms of both accuracy and computational cost, particularly in scenarios with rapid phase changes. The proposed method is therefore well suited for simulating sudden large changes in phase field models in both scientific and engineering applications.

1. Introduction

This study proposes an adaptive time-stepping algorithm for the AC equation [1]:

$$\frac{\partial \phi(\mathbf{x}, t)}{\partial t} = -\frac{F'(\phi(\mathbf{x}, t))}{\epsilon^2} + \Delta \phi(\mathbf{x}, t), \quad (1)$$

where $\phi(\mathbf{x}, t)$ represents the order parameter for the phase-field function in a region $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$), $F(\phi(\mathbf{x}, t)) = 0.25(\phi(\mathbf{x}, t)^2 - 1)^2$, ϵ is a small positive constant. We consider the zero Neumann boundary condition. The proposed scheme adapts the time step according to the solution behavior measured in the finite difference setting, and provides substantial improvements in accuracy and computational cost compared with fixed time-stepping approaches.

Phase separation and interface dynamics are phenomena commonly encountered in scientific and engineering contexts, often modeled and applied using the Allen–Cahn (AC) type equations [2–7]. These equations have been widely applied to describe tissue growth [8], surface reconstruction [9], data assimilation [10], and shape transformation [11,12] in complex materials and physical systems [13,14]. Through these applications, the AC equation captures the interplay between interfacial energy and diffusion, which governs the temporal evolution of phase boundaries. Such dynamics often involve rapid phase transitions and localized interface motion, which impose stringent requirements on the temporal step of numerical simulations [15]. Therefore, efficient numerical methods are required to accurately resolve these fast transient behaviors while maintaining computational efficiency [16–18]. An adaptive time-stepping scheme for the AC equation was developed using the Runge-Kutta-Fehlberg method to efficiently control multiscale dynamics [19]. Two second-order stable schemes were developed for the fractional-in-space AC equation. Crank-Nicolson

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and BDF2 methods were used for the temporal discretization [20]. An adaptive time-stepping method was developed to accelerate the computation [20,21]. Efficient second-order in time approximations were proposed for AC and CH equations. To reduce computational expenses, the authors developed an adaptive time-stepping algorithm that uses numerical dissipation, as outlined by a discrete energy law at each time step, as described in [22]. The Runge-Kutta methods were extended to automatically select the time step to control the error [23]. An adaptive control time step for the updated model method in neutron diffusion adjusts the time step automatically based on error detection [24]. An improved adaptive time step algorithm for natural gas pipeline network simulations reduces trial calculations and improves computational efficiency by introducing the concept of energy number to assess boundary condition variations [25]. Time-stepping schemes were developed and these schemes can use different time step sizes for each of the subproblems. A posteriori error estimator was developed for an adaptive time-stepping process [26].

Several adaptive time step strategies for the AC equation have been reported in the literature, for example in [19,22]. These approaches usually employ multi-step time discretizations, and their time step controllers depend on solution values at two or more previous time levels. To the best of our knowledge, the literature does not provide a one-step adaptive method that matches the structure of the present scheme. Therefore, a direct comparison of error versus computational cost with [19,22] would mix the effects of the multi-step discretization order, the start-up procedure, and the controller design, and it would not isolate the contribution of one-step adaptivity. The present work resolves a complementary objective, which asks whether a one-step adaptive method can provide both efficiency and accuracy for the AC equation within a one-step scheme. For this reason, the numerical section compares the proposed method with the explicit Euler scheme with a fixed uniform time step, which serves as a standard one-step baseline. This choice isolates the impact of time step adaptivity within the same one-step class under the same spatial discretization and test problems.

To position the proposed adaptive explicit Euler scheme within the broader landscape of partial differential equation time integrators, the numerical test section provides benchmark comparisons with several representative alternatives. These alternatives reflect common design choices in time integration for AC type problems. The goal of the benchmark is to quantify the trade-off among accuracy, computational cost, and implementation complexity under a consistent spatial discretization and identical boundary treatment.

The proposed adaptive time-stepping scheme has several advantages. First, the update is fully explicit and local, and it avoids nonlinear solvers and iterative procedures. Second, the adaptive step size rule constrains the maximum one-step update in the L^∞ -norm. As a result, the rule selects smaller time steps during rapid phase changes and larger time steps during slow evolution, which improves efficiency for multiscale dynamics. Third, the algorithm has a straightforward algorithmic structure, and the manuscript provides Algorithms 1–2 and MATLAB source code to facilitate reproducibility and implementation. Its limitations are first-order temporal accuracy with a stability restricted $\Delta t_{\max}$, and the absence of a classical local error estimator or an unconditional discrete energy law.

This paper is organized as follows. Section 2 describes the proposed algorithm for computing the numerical solution. Section 3 presents the numerical experiments and their results. Section 4 concludes the article.

2. Numerical methods

2.1. Two-dimensional space

We discretize the domain $\Omega = (L_x, R_x) \times (L_y, R_y)$ using a cell-centered grid $\Omega_h = \{(x_i, y_j) | x_i = L_x + (i - 0.5)h, 1 \leq i \leq N_x \text{ and } y_j = L_y + (j - 0.5)h, 1 \leq j \leq N_y\}$, where $h = (R_x - L_x)/N_x$ is the spatial step size, and N_x is the number of spatial points (Fig. 1). The cell-centered layout aligns with the midpoint rule for cell averages, which attains second-order accuracy on uniform meshes. This choice also provides a symmetric discretization of the diffusion operator. Centered flux differences across cell faces produce the standard five-point Laplacian stencil, and Taylor expansion yields an $O(h^2)$ spatial truncation error for smooth solutions. In addition, the first unknown lies at a distance $h/2$ from the boundary. This property allows a symmetric mirror extension with ghost cells, so the scheme enforces the homogeneous Neumann boundary condition while it retains the centered stencil near the boundary and avoids one-sided differences. A consistent implementation with ghost layers appears in the Appendix source code.

The numerical approximation of the solution is defined by

$$\phi_{ij}^n = \phi(x_i, y_j, t^n),$$

where Δt^n is the nonuniform temporal step size, $t^0 = 0$, $t^n = t^{n-1} + \Delta t^n$, for $n \geq 1$. Let $\mu_{ij}^n = \frac{(\phi_{ij}^n)^3 - \phi_{ij}^n}{\epsilon^2} - \Delta_h \phi_{ij}^n$, where the discrete Laplace operator in 2D is defined as $\Delta_h \phi_{ij}^n = (\phi_{i-1,j}^n + \phi_{i+1,j}^n + \phi_{i,j-1}^n + \phi_{i,j+1}^n - 4\phi_{ij}^n)/h^2$. The explicit Euler scheme is applied to Eq. (1):

$$\frac{\phi_{ij}^{n+1} - \phi_{ij}^n}{\Delta t^{n+1}} = -\mu_{ij}^n. \quad (2)$$

The zero Neumann boundary condition is considered for $n \geq 0$. In 2D, we describe an adaptive time-stepping method. The numerical solution's maximum displacement within a single time step is constrained by a tolerance tol . Both sides of Eq. (2) are then scaled by Δt^{n+1} , as

$$\phi_{ij}^{n+1} - \phi_{ij}^n = -\Delta t^{n+1} \mu_{ij}^n, \quad 1 \leq i \leq N_x, 1 \leq j \leq N_y. \quad (3)$$

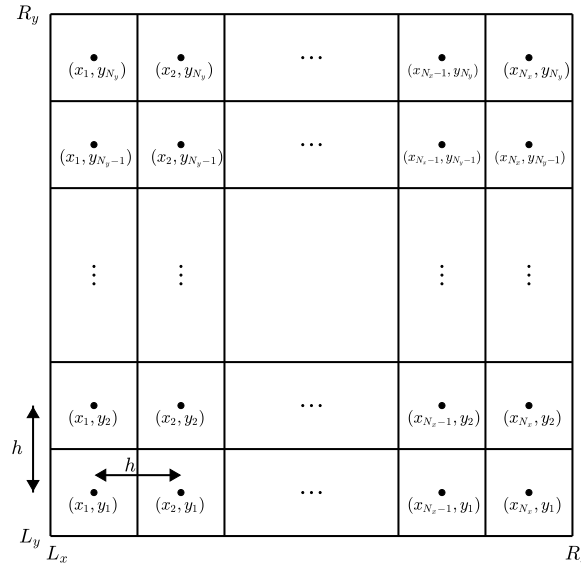


Fig. 1. A cell-centered computational mesh in 2D.

We define the maximum norm in 2D as

$$\|\phi\|_\infty = \max_{\substack{1 \leq i \leq N_x \\ 1 \leq j \leq N_y}} |\phi_{ij}|. \quad (4)$$

Then, from Eq. (3), it is necessary for the following condition to be met:

$$\|\phi^{n+1} - \phi^n\|_\infty = \Delta t^{n+1} \|\mu^n\|_\infty \leq \text{tol}. \quad (5)$$

From Eq. (5), the constraint of the temporal step sizes is obtained:

$$\Delta t^{n+1} \leq \text{tol} / \|\mu^n\|_\infty. \quad (6)$$

Then, we have

$$\Delta t^{n+1} = s \times \min \left(\text{tol} / \|\mu^n\|_\infty, \Delta t_{\max} \right), \quad (7)$$

where s is a safety factor and $\Delta t_{\max} = \epsilon^2 h^2 / (2h^2 + 4\epsilon^2)$ is a given maximum temporal step size [27]. The safety factor $s \in (0, 1)$ provides a strict margin for the time step restriction that stems from the maximum-norm constraint and the stability cap $\Delta t_{\max}$. In our implementation, we set $s = 0.99$ which maintains a one-percent margin while it keeps Δt^{n+1} close to the admissible upper bound. A smaller value of s reduces Δt^{n+1} and increases the number of steps, and a value closer to 1 reduces the number of steps but may reduce the margin. Therefore, s offers a simple cost robustness trade-off, and we adopt $s = 0.99$ as a conservative default.

To motivate the stability cap $\Delta t_{\max}$ used in the adaptive rule, we briefly outline the maximum principle argument for the explicit Euler update. Consider the fully explicit scheme $\phi^{n+1} = \phi^n - \Delta t \left(((\phi^n)^3 - \phi^n) / \epsilon^2 - \Delta_h \phi^n \right)$ with the standard $(2d + 1)$ -point discrete

Laplacian $\Delta_h \phi_p = \left(\sum_{q \in \mathcal{N}(p)} \phi_q - 2d \phi_p \right) / h^2$, where $|\mathcal{N}(p)| = 2d$ and d is the dimensionality. Assume $|\phi_q| \leq 1$ for all grid points, so $\sum_{q \in \mathcal{N}(p)} \phi_q \in [-2d, 2d]$. A case-by-case estimate for $\phi_p^n \in [-1, 1]$ yields sufficient conditions that enforce $\phi_p^{n+1} \in [-1, 1]$, in particular $\Delta t \leq h^2 / (2d)$ for the pure diffusion contribution and a sharper bound that couples reaction and diffusion for $0 < \phi_p^n < 1$. Using $0 < \phi(1 + \phi) < 2$ on $[-1, 1]$, the latter condition reduces to the dimension-dependent cap $\Delta t_{\max} = \epsilon^2 h^2 / (2h^2 + 2d\epsilon^2)$. Therefore, $\Delta t_{\max}^{2D} = \epsilon^2 h^2 / (2h^2 + 4\epsilon^2)$ and $\Delta t_{\max}^{3D} = \epsilon^2 h^2 / (2h^2 + 6\epsilon^2)$, where the factors 4 and 6 correspond to the number of nearest neighbors in the discrete Laplacian [27].

From Eq. (3), the next time solution is obtained for $1 \leq i \leq N_x$ and $1 \leq j \leq N_y$:

$$\phi_{ij}^{n+1} = \phi_{ij}^n - \Delta t^{n+1} \mu_{ij}^n. \quad (8)$$

Continue this procedure as long as $t^n + \Delta t^{n+1} \leq T$ to determine the numerical value of $\phi(x, y, T)$. For adaptive time steps, it is possible for t^{n+1} to surpass the target end time. To prevent this, $t^{n+1} = t^n + \Delta t^{n+1} > T$, adjust Δt^{n+1} to $T - t^n$.

Remark 1. In Algorithms 1–2, if $t^n + \Delta t^{n+1} > T$, we set $\Delta t^{n+1} := T - t^n$ so that the numerical solution is reported exactly at the final time T . This modification does not violate the adaptivity constraint, because it only decreases the step size. Indeed, the adaptive rule selects Δt^{n+1} as an admissible upper bound subject to the update constraint and the stability cap; replacing it by $\Delta t_{\text{final}}^{n+1} = T - t^n < \Delta t^{n+1}$ yields

$$\|\phi^{n+1} - \phi^n\|_\infty = \Delta t_{\text{final}}^{n+1} \|\mu^n\|_\infty \leq \Delta t^{n+1} \|\mu^n\|_\infty \leq s \text{tol}.$$

This adjustment affects only the last step and reduces $\max_k \Delta t^k$ or leaves it unchanged. Therefore, it does not degrade the expected global temporal accuracy of the explicit Euler discretization, and it can improve robustness near T .

A comprehensive computational algorithm for each stage of the proposed adaptive time-stepping method is outlined in Algorithm 1.

Algorithm 1 Adaptive time-stepping scheme for the AC equation in 2D.

INPUT endpoints L_x, R_x, L_y, R_y ; numbers of grid points N_x, N_y ; safety factor s ; initial condition; final time T ; tolerance tol ; maximum size of time step Δt_{max} .

OUTPUT approximation of $\phi(x, y, T)$.

Step 1 Initial setting

Set $h = (R_x - L_x)/N_x$, $n = 0$, and $t = 0$

For $i = 1, \dots, N_x$ do

For $j = 1, \dots, N_y$ do

$x_i = L_x + (i - 0.5)h$, $y_j = L_y + (j - 0.5)h$

$\phi_{ij}^0 = \phi(x_i, y_j, 0)$

Step 2 While ($t < T$), do Steps 3–5.

Step 3 For $i = 1, \dots, N_x$ do

For $j = 1, \dots, N_y$ do

$$\mu_{ij}^n = \frac{(\phi_{ij}^n)^3 - \phi_{ij}^n}{\epsilon^2} - \Delta_h \phi_{ij}^n$$

Step 4 Set $\Delta t^{n+1} = s \min \left(\text{tol} / \|\mu^n\|_\infty, \Delta t_{\text{max}} \right)$

If $t + \Delta t^{n+1} > T$, then $\Delta t^{n+1} = T - t$.

Step 5 Set $\phi_{ij}^{n+1} = \phi_{ij}^n - \Delta t^{n+1} \mu_{ij}^n$

$t = t + \Delta t^{n+1}$ and $n = n + 1$.

with homogeneous Neumann boundary condition

Step 6 OUTPUT approximation of $\phi(x, y, T)$

STOP.

Remark 2. The proposed scheme treats the AC nonlinearity in a fully explicit manner. At each time step, the cubic reaction term $(\phi^3 - \phi)/\epsilon^2$ is evaluated at time level t^n pointwise on the cell-centered grid and is combined with the discrete diffusion term. In

particular, we define $\mu_{ij}^n = \frac{(\phi_{ij}^n)^3 - \phi_{ij}^n}{\epsilon^2} - \Delta_h \phi_{ij}^n$, and the update takes the explicit form $\phi_{ij}^{n+1} = \phi_{ij}^n - \Delta t^{n+1} \mu_{ij}^n$. This choice avoids any nonlinear algebraic system and removes the need for Newton-type iterations, since the nonlinearity is local at each grid point. The adaptive time step rule uses $\|\mu^n\|_\infty$, so the method reduces Δt^{n+1} when the reaction term dominates and permits larger Δt^{n+1} when the solution evolves slowly.

Remark 3. The adaptive rule in Eqs. (5)–(7) constrains the maximum one-step update in the L^∞ -norm: $\|\phi^{n+1} - \phi^n\|_\infty = \Delta t^{n+1} \|\mu^n\|_\infty \leq s \text{tol}$, and $\mu^n = \frac{(\phi^n)^3 - \phi^n}{\epsilon^2} - \Delta_h \phi^n$. Hence, tol controls the stepwise variation of the phase-field variable and determines the time step through $\Delta t^{n+1} \propto \text{tol} / \|\mu^n\|_\infty$ up to the stability cap Δt_{max} . We emphasize that this criterion does not provide an embedded local truncation error estimator. For the explicit Euler method, the temporal discretization is first order, and standard stability theory for one-step methods yields a global temporal error bound of the form $\max_n \|e^n\| \leq C \max_n \Delta t^n$ under usual Lipschitz and regularity assumptions. Therefore, in smooth regimes, a smaller tol leads to a smaller $\max_n \Delta t^n$ and reduces the expected global temporal error proportionally, until Δt^{n+1} reaches the imposed bound Δt_{max} . For classical local-error control, one may use step-doubling or embedded RK pairs; see, e.g., standard references on adaptive time-stepping [28,29].

2.2. Three-dimensional space

We present a discretization for the given 3D domain $\Omega = (x_L, x_R) \times (y_L, y_R) \times (z_L, z_R)$ as $\Omega_h = \{(x_i, y_j, z_k) | x_i = x_L + (i - 0.5)h, i = 1, \dots, N_x, y_j = y_L + (j - 0.5)h, j = 1, \dots, N_y \text{ and } z_k = z_L + (k - 0.5)h, k = 1, \dots, N_z\}$, where $h = (x_R - x_L)/N_x$ is a spatial step size and N_x is the number of spatial points (see, Fig. 2).

For simplicity, we denote the notation as

$$\phi_{ijk}^n = \phi(x_i, y_j, z_k, t^n),$$

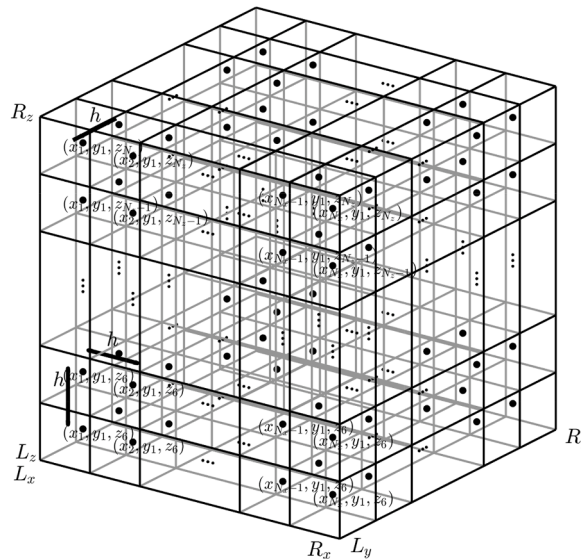


Fig. 2. A cell-centered computational mesh in 3D.

We apply the explicit Euler method for $i = 1, \dots, N_x$, $j = 1, \dots, N_y$, $k = 1, \dots, N_z$, and $n = 0, 1, \dots, N_t$.

$$\frac{\phi_{ijk}^{n+1} - \phi_{ijk}^n}{\Delta t^{n+1}} = -\mu_{ijk}^n, \quad (9)$$

where $\mu_{ijk}^n = \frac{(\phi_{ijk}^n)^3 - \phi_{ijk}^n}{\epsilon^2} - \Delta_h \phi_{ijk}^n$ and the discrete Laplace operator is defined by $\Delta_h \phi_{ijk}^n = (\phi_{i-1,j,k}^n + \phi_{i+1,j,k}^n + \phi_{i,j-1,k}^n + \phi_{i,j+1,k}^n + \phi_{i,j,k-1}^n + \phi_{i,j,k+1}^n - 6\phi_{ijk}^n)/h^2$. The homogeneous Neumann boundary condition is applied for $n \geq 0$.

In 3D, we describe an adaptive time-stepping method. The numerical solution's maximum displacement within a single time step is constrained by the tolerance tol . Both sides of Eq. (9) are then scaled by Δt^{n+1} , for $1 \leq i \leq N_x$, $1 \leq j \leq N_y$, $1 \leq k \leq N_z$:

$$\phi_{ijk}^{n+1} - \phi_{ijk}^n = -\Delta t^{n+1} \mu_{ijk}^n, \quad (10)$$

We define a maximum norm in 3D as

$$\|\phi\|_\infty = \max_{\substack{1 \leq i \leq N_x \\ 1 \leq j \leq N_y \\ 1 \leq k \leq N_z}} |\phi_{ijk}|. \quad (11)$$

Subsequently, as dictated by Eq. (10), the ensuing condition must be fulfilled:

$$\|\phi^{n+1} - \phi^n\|_\infty = \Delta t^{n+1} \|\mu^n\|_\infty \leq tol. \quad (12)$$

Based on Eq. (12), we establish the limitations on the temporal step sizes:

$$\Delta t^{n+1} \leq tol / \|\mu^n\|_\infty. \quad (13)$$

Then, we consider

$$\Delta t^{n+1} = s \times \min \left(tol / \|\mu^n\|_\infty, \Delta t_{\max} \right), \quad (14)$$

where s is a safety factor and $\Delta t_{\max} = \epsilon^2 h^2 / (2h^2 + 6\epsilon^2)$ is a given maximum temporal step size [27]. From Eq. (10), we have the next time solution for $1 \leq i \leq N_x$, $1 \leq j \leq N_y$ and $1 \leq k \leq N_z$:

$$\phi_{ijk}^{n+1} = \phi_{ijk}^n - \Delta t^{n+1} \mu_{ijk}^n. \quad (15)$$

Remark 4. In Eqs. (9)-(15), the cubic reaction term $(\phi_{ijk}^n)^3 - \phi_{ijk}^n$ is computed explicitly at time level t^n and combined with the 7-point discrete Laplacian $\Delta_h \phi_{ijk}^n$ in the definition of μ_{ijk}^n . Therefore, the update $\phi_{ijk}^{n+1} = \phi_{ijk}^n - \Delta t^{n+1} \mu_{ijk}^n$ proceeds by a pointwise evaluation and does not require any nonlinear iteration; the time step rule based on $\|\mu^n\|_\infty$ reduces Δt_{n+1} when the reaction term dominates.

A detailed 3D numerical method for each step of the proposed algorithm is described in Algorithm 2.

The adaptive time step selection is based on the maximum norm between ϕ^{n+1} and ϕ^n , set to keep the largest movement within an established tolerance limit. This criterion adjusts the time step downward during fast changes and increases it during slower periods.

Algorithm 2 Adaptive time-stepping scheme for the AC equation in 3D.

INPUT endpoints $x_L, x_R, y_L, y_R, z_L, z_R$; number of grid points N_x, N_y, N_z ; safety factor s ; initial condition; final time T ; tolerance tol ; maximum size of time step $\Delta t_{\max}$.

OUTPUT approximation of $\phi(x, y, z, T)$.

Step 1 Initial setting

Set $h = (x_R - x_L)/N_x$, $n = 0$, and $t = 0$

For $i = 1, \dots, N_x$ do

For $j = 1, \dots, N_y$ do

For $k = 1, \dots, N_z$ do

$x_i = x_L + (i - 0.5)h$, $y_j = y_L + (j - 0.5)h$, $z_k = z_L + (k - 0.5)h$

$\phi_{ijk}^0 = \phi(x_i, y_j, z_k, 0)$

Step 2 While ($t < T$), do Steps 3–5.**Step 3** For $i = 1, \dots, N_x$ do

For $j = 1, \dots, N_y$ do

For $k = 1, \dots, N_z$ do

$$\mu_{ijk}^n = \frac{(\phi_{ijk}^n)^3 - \phi_{ijk}^n}{\epsilon^2} - \Delta_h \phi_{ijk}^n$$

Step 4 Set $\Delta t^{n+1} = s \min \left(tol / \|\mu^n\|_\infty, \Delta t_{\max} \right)$

If $t + \Delta t^{n+1} > T$, then $\Delta t^{n+1} = T - t$.

Step 5 Set $\phi_{ijk}^{n+1} = \phi_{ijk}^n - \Delta t^{n+1} \mu_{ijk}^n$.

$t = t + \Delta t^{n+1}$ and $n = n + 1$.

with homogeneous Neumann boundary condition

Step 6 OUTPUT approximation of $\phi(x, y, z, T)$

STOP.

This method of time adaptivity, as described in our updating algorithm, is efficient and straightforward, and it avoids the need to recalculate numerical solutions that do not meet specific conditions, unlike the approach described in [19]. Our update procedure is implemented as a one-step process.

Remark 5. The same interpretation applies in 3D: the adaptive choice constrains $\|\phi^{n+1} - \phi^n\|_\infty = \Delta t^{n+1} \|\mu^n\|_\infty$ with $\mu^n = ((\phi^n)^3 - \phi^n)/\epsilon^2 - \Delta_h \phi^n$. Since the explicit Euler discretization is first order in time, standard one-step stability theory implies that a smaller tol reduces $\max_n \Delta t^n$ and decreases the expected global temporal error in smooth regimes.

2.3. Consistency and error discussion

We add a brief discussion on the accuracy of the proposed fully discrete scheme. In both 2D and 3D, the diffusion operator is approximated by the standard second-order central finite difference Laplacian, and the time discretization uses the explicit Euler method with a nonuniform time step Δt^{n+1} ; see Eq. (2) and Algorithm 1. For a sufficiently smooth solution, a Taylor expansion in time gives

$$\frac{\phi(\mathbf{x}_{ij}, t_{n+1}) - \phi(\mathbf{x}_{ij}, t_n)}{\Delta t^{n+1}} = \partial_t \phi(\mathbf{x}_{ij}, t_n) + \mathcal{O}(\Delta t^{n+1}), \quad (16)$$

and the central finite difference Laplacian satisfies

$$\Delta \phi(\mathbf{x}_{ij}, t_n) = \Delta_h \phi(\mathbf{x}_{ij}, t_n) + \mathcal{O}(h^2). \quad (17)$$

Hence, the local truncation error satisfies

$$\tau_{ij}^{n+1} = \mathcal{O}(\Delta t^{n+1}) + \mathcal{O}(h^2), \quad (18)$$

which implies first-order consistency in time and second-order consistency in space. Under standard stability and regularity assumptions, the global discretization error typically scales as $\mathcal{O}(\Delta t_{\max}) + \mathcal{O}(h^2)$. We also clarify the role of the adaptive step-size rule. Let

$$\mu^n := \frac{(\phi^n)^3 - \phi^n}{\epsilon^2} - \Delta_h \phi^n, \quad \phi^{n+1} = \phi^n - \Delta t^{n+1} \mu^n. \quad (19)$$

The adaptive strategy enforces an update constraint of the form $\|\phi^{n+1} - \phi^n\|_\infty \leq s \, tol$ (subject to the imposed stability bound), which yields $\Delta t^{n+1} \propto tol / \|\mu^n\|_\infty$. This criterion reduces Δt^{n+1} when $\|\mu^n\|_\infty$ becomes large and allows a larger step when the solution changes slowly. We note that this mechanism targets a bounded update per step; it does not provide a classical embedded local-error estimator.

2.4. Convergence analysis

We provide a formal convergence analysis for the variable-step explicit Euler scheme used in this study. Let the semi-discrete problem derived from the spatial discretization be denoted by the system of ordinary differential equations:

$$\frac{d\Phi(t)}{dt} = F_h(\Phi(t)), \quad \Phi(0) = \Phi^0, \quad (20)$$

where $\Phi(t) \in \mathbb{R}^N$ is the vector of exact semi-discrete solutions at grid points, and $F_h : \mathbb{R}^N \rightarrow \mathbb{R}^N$ represents the discrete reaction–diffusion operator. The proposed adaptive scheme generates the numerical approximation $\Phi^n \approx \Phi(t_n)$ via the update rule:

$$\Phi^{n+1} = \Phi^n + \Delta t^{n+1} F_h(\Phi^n), \quad (21)$$

where the time step Δt^{n+1} is dynamically determined by the adaptive criterion subject to the stability cap $\Delta t_{\max}$.

Assumptions. To establish the convergence of the scheme, we make the following standard assumptions:

1. The exact semi-discrete solution $\Phi(t)$ is twice continuously differentiable with respect to time on the interval $[0, T]$. We define the bound on the second derivative as $M := \max_{t \in [0, T]} \|\Phi''(t)\|_\infty$.
2. Based on the discrete maximum principle preserved by the stability condition (see [27]), we assume that the numerical solution remains within a bounded domain D (e.g., $[-1, 1]^N$). Since the cubic reaction term in the AC equation is smooth, the operator F_h satisfies the Lipschitz condition on D with a constant L_h :

$$\|F_h(U) - F_h(V)\|_\infty \leq L_h \|U - V\|_\infty, \quad \forall U, V \in D. \quad (22)$$

Error Analysis. Let $e^n := \Phi(t_n) - \Phi^n$ denote the global error at time t_n . Applying a Taylor expansion to the exact solution around t_n , we obtain:

$$\Phi(t_{n+1}) = \Phi(t_n) + \Delta t^{n+1} F_h(\Phi(t_n)) + \tau^{n+1}, \quad (23)$$

where the local truncation error τ^{n+1} satisfies the bound $\|\tau^{n+1}\|_\infty \leq \frac{1}{2} M (\Delta t^{n+1})^2$. Subtracting the numerical scheme from the exact expansion yields the error recurrence relation:

$$e^{n+1} = e^n + \Delta t^{n+1} (F_h(\Phi(t_n)) - F_h(\Phi^n)) + \tau^{n+1}.$$

Taking the L^∞ -norm and applying the Lipschitz condition (22), we obtain:

$$\|e^{n+1}\|_\infty \leq (1 + L_h \Delta t^{n+1}) \|e^n\|_\infty + \frac{1}{2} M (\Delta t^{n+1})^2. \quad (24)$$

Assuming $e^0 = 0$, we apply the discrete Grönwall inequality. The global error at step n is bounded by:

$$\begin{aligned} \|e^n\|_\infty &\leq \sum_{k=0}^{n-1} \left(\prod_{j=k+1}^{n-1} (1 + L_h \Delta t^{j+1}) \right) \frac{M}{2} (\Delta t^{k+1})^2 \\ &\leq \exp \left(L_h \sum_{j=1}^n \Delta t^j \right) \sum_{k=0}^{n-1} \frac{M}{2} (\Delta t^{k+1})^2. \end{aligned} \quad (25)$$

Using the relation $\sum_{j=1}^n \Delta t^j = t_n \leq T$ and the estimate $(\Delta t^{k+1})^2 \leq \left(\max_{1 \leq m \leq N_T} \Delta t^m \right) \Delta t^{k+1}$, we derive the final convergence bound:

$$\begin{aligned} \|e^n\|_\infty &\leq e^{L_h T} \frac{M}{2} \left(\max_{1 \leq m \leq N_T} \Delta t^m \right) \sum_{k=0}^{n-1} \Delta t^{k+1} \\ &\leq \left(\frac{TM}{2} e^{L_h T} \right) \max_{1 \leq k \leq N_T} \Delta t^k. \end{aligned} \quad (26)$$

Thus, the global error satisfies

$$\max_{0 \leq n \leq N_T} \|e^n\|_\infty \leq \tilde{C} \max_{1 \leq k \leq N_T} \Delta t^k, \quad \text{with } \tilde{C} = \frac{TM}{2} e^{L_h T}. \quad (27)$$

This result explicitly confirms that the proposed scheme maintains first-order temporal convergence as $\max_k \Delta t^k \rightarrow 0$ with a fixed h .

Remark 6. We relate this theoretical result to the proposed adaptive strategy. The adaptive criterion enforces a bound on the stepwise update magnitude:

$$\|\Phi^{n+1} - \Phi^n\|_\infty = \Delta t^{n+1} \|F_h(\Phi^n)\|_\infty \leq s \, tol, \quad (28)$$

up to the imposed stability cap $\Delta t_{\max}$. While this rule does not provide an embedded local truncation error estimator in the classical sense, it effectively links the tolerance parameter tol to the time step size via $\Delta t_{n+1} \propto tol / \|F_h(\Phi^n)\|_\infty$. Consequently, a smaller tolerance tol typically reduces the maximum time step size $\max_k \Delta t^k$ in regimes where the solution varies smoothly, thereby decreasing the expected global temporal error bound derived above.

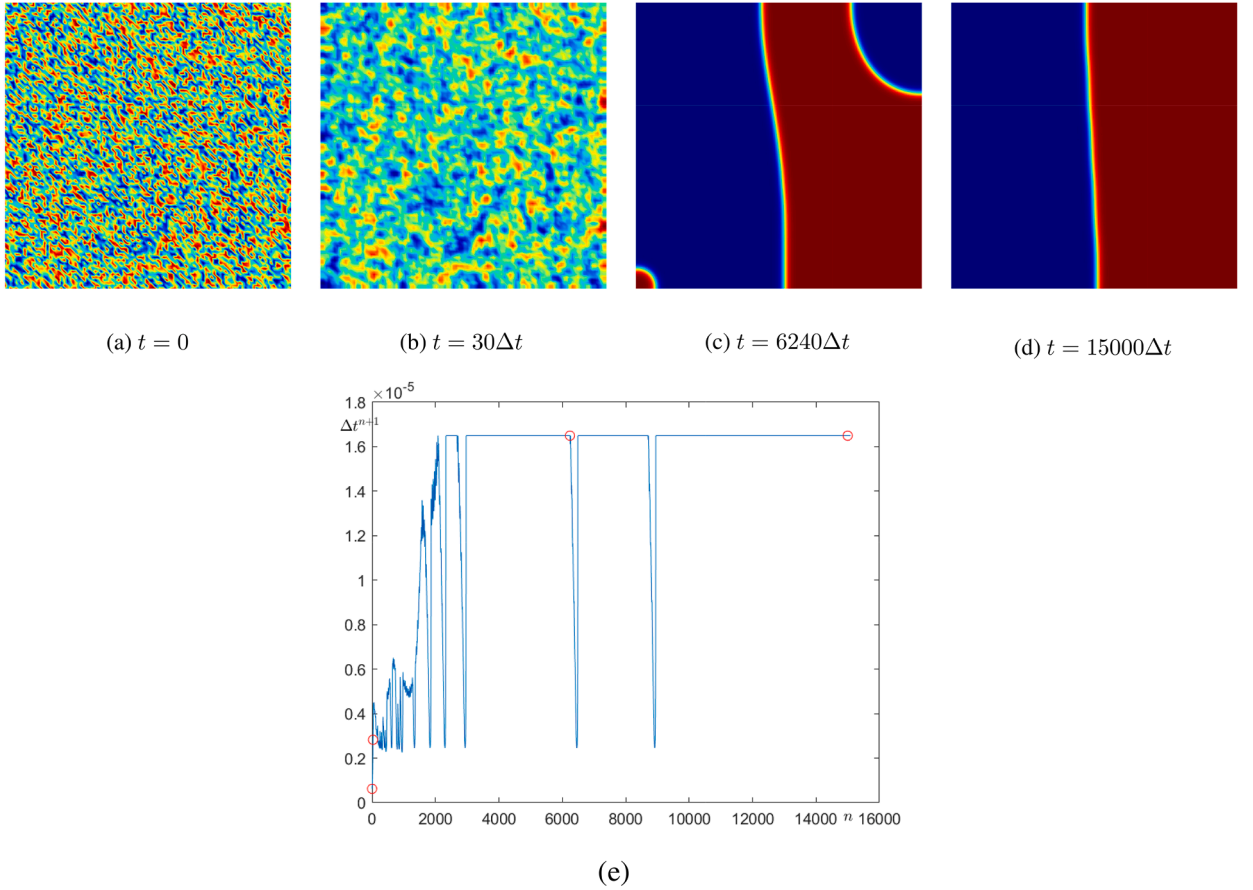


Fig. 3. (a)-(d) snapshots of the temporal evolution of phase separation at $t = 0, 30\Delta t, 6240\Delta t, 15000\Delta t$, respectively. (e) the changes in the time step size Δt^{n+1} .

3. Computational experiments

In this section, we evaluate accuracy and efficiency through the relation between numerical error and computational cost. We report the cost by the number of accepted time steps. We compare two one-step schemes: the proposed adaptive method and the explicit Euler method with a fixed time step. This baseline provides a direct one-step reference and avoids structural differences that arise in multi-step adaptive schemes, such as the need for solution history and a start-up stage, as well as a different per-step cost model. The reported results therefore quantify the advantage of one-step adaptivity in a controlled setting.

3.1. Two-dimensional space

3.1.1. Phase separation

In Fig. 3, we investigate the phase separation process of the AC equation over time. The parameter and initial condition of Fig. 3 are defined as $\Omega = (0, 1) \times (0, 1)$ with 100×100 mesh, where $h = 1/N_x$, $\epsilon = h$, $tol = 2.0e - 2$, $\Delta t_{\max} = \frac{\epsilon^2 h^2}{2h^2 + 4\epsilon^2}$, $T = 0.22$, and the initial condition is defined by $\phi_{ij}^0 = 0.5(2\text{rand}(x_i, y_j) - 1)$, where $\text{rand}(x_i, y_j)$ are random variables uniformly distributed on $(0, 1)$. The numerical solution is stored on an $(N_x + 2) \times (N_y + 2)$ array that includes one layer of ghost points. The zero Neumann boundary condition is imposed by mirror extension on the ghost layers: $\phi_{1,j}^n = \phi_{2,j}^n$, $\phi_{N_x+2,j}^n = \phi_{N_x+1,j}^n$, $\phi_{i,1}^n = \phi_{i,2}^n$, and $\phi_{i,N_y+2}^n = \phi_{i,N_y+1}^n$ for all $n \geq 0$.

In Fig. 3(a), the initial condition shows randomly assigned field values. The blue and yellow regions represent different phase states (Fig. 3(b)). The boundary between the blue and yellow regions becomes more distinct (Fig. 3(c)). In Fig. 3(d), the movement of the interface nearly stops, forming a simplified structure at the boundary between the blue and yellow regions. Fig. 3(e) shows the change in the time step size Δt^{n+1} . As phase separation progresses, the interface is smoothed, and the phase structure becomes clearer.

3.1.2. Motion by mean curvature

In Fig. 4, we study motion by mean curvature using the proposed method. We consider the square domain $\Omega = (0, 1) \times (0, 1)$ on 100×100 mesh. We set $N_x = 100$, $N_y = 100$, $h = 1/N_x$, $tol = 1.0e - 3$, $\epsilon = h$, $\Delta t_{\max} = \epsilon^2 h^2 / (2h^2 + 4\epsilon^2)$, and $T = 0.1$. In Fig. 4(a), the initial condition consists of four squares. As time evolves, the squares approach circular shapes and shrink; see Fig. 4(b). The smallest square disappears first, and the remaining ones disappear in sequence; see Fig. 4(c). After that, the last circle continues to shrink and vanishes; see Fig. 4(d). Fig. 4(e) reports the evolution of the time-step size Δt^{n+1} . The time step attains larger values when the interface evolves slowly (for example, when the normal velocity is small and the curvature is low), and it decreases during rapid shrinkage.

The AC equation admits the free-energy functional

$$\mathcal{E}(\phi) = \int_{\Omega} \left(\frac{1}{4\epsilon^2} (\phi^2 - 1)^2 + \frac{1}{2} |\nabla \phi|^2 \right) dx, \quad (29)$$

and the continuous gradient flow structure implies $dE(\phi(t))/dt \leq 0$ for smooth solutions. To report the energy decay in Fig. 4(f), we define the discrete energy

$$\mathcal{E}_h(\phi^n) = h^2 \sum_{(i,j) \in \Omega_h} \left(\frac{1}{4\epsilon^2} ((\phi_{ij}^n)^2 - 1)^2 + \frac{1}{2} \sum_{m=1}^2 (D_m^+ \phi_{ij}^n)^2 \right), \quad (30)$$

where D_m^+ denotes the first-order forward difference operator in the m -th coordinate direction,

$$(D_m^+ \phi)_{ij} = \frac{\phi_{i+\mathbf{e}_m, j} - \phi_{ij}}{h} \quad (m=1), \quad (D_m^+ \phi)_{ij} = \frac{\phi_{i, j+\mathbf{e}_m} - \phi_{ij}}{h} \quad (m=2),$$

and $\mathbf{e}_m$ denotes the unit index vector in the m -th direction. The homogeneous Neumann boundary condition is imposed by ghost-cell reflection, so these difference operators remain consistent near the boundary.

Fig. 4(f) shows the discrete energy decay and the evolution of the maximum and minimum values of ϕ^n over time. The energy is non-increasing in this test, and $\max \phi^n$ and $\min \phi^n$ remain within the prescribed bounds, which supports the discrete maximum principle. We note that the explicit Euler discretization does not guarantee an unconditional discrete energy law for arbitrary step sizes. Nevertheless, the explicit update admits a gradient descent interpretation for a discrete energy functional, and a standard Taylor-type descent estimate gives a sufficient condition for stepwise energy decrease. In the 1D setting, such an argument yields $\mathcal{E}_h(\phi^{n+1}) \leq \mathcal{E}_h(\phi^n)$ under a time-step restriction [27] of the form $\Delta t \leq \epsilon^2 h^2 / (2h^2 + 2\epsilon^2)$. The same mechanism extends to higher dimensions after one replaces the diffusion contribution by the corresponding bound associated with $2d$ nearest neighbors, which motivates the practical cap $\Delta t_{\max} = \epsilon^2 h^2 / (2h^2 + 2d\epsilon^2)$. The adaptive controller imposes $\Delta t^{n+1} \leq s \Delta t_{\max}$ with $s = 0.99$, and Fig. 4(f) exhibits a non-increasing discrete energy profile under this restriction. We view this behavior as a step-size-dependent property of the explicit method rather than an unconditional energy-stability guarantee.

3.1.3. Shrinking circle benchmark

In Fig. 5(a), the analytic radius in two dimensions at time t is given as $R(t) = \sqrt{r^2 - 2t}$ and the initial condition of numerical radius is defined as

$$\phi(x, y, 0) = \tanh \left(\frac{r - \sqrt{(x-0.5)^2 + (y-0.5)^2}}{\sqrt{2}\epsilon} \right)$$

on $\Omega = (-0.5, 1.5) \times (-0.5, 1.5)$ with 100×100 mesh, where $r = 0.7$, $h = 2/N_x$, $\epsilon = 4h$, $tol = 4.0e - 3$, $\Delta t_{\max} = \frac{\epsilon^2 h^2}{2h^2 + 4\epsilon^2}$.

Fig. 5(a) compares the analytic radius $R(t)$ with numerical radii obtained for three choices of parameter s . The analytic curve decreases from $R(0) = 0.7$ to $R(T) = 0$ at $T = 0.245$, which indicates finite time extinction. All cases exhibit near coincidence with the analytic law over the full interval, and it reproduces the extinction time within the plot resolution. Figs. 5(b)–(d) present the step size sequence Δt^{n+1} versus the iteration index n for $s = 0.6, 0.8$ and 0.99 . In all runs, Δt stays near a constant level for most of the simulation, then drops sharply near the terminal stage. For $s = 0.6$, Δt stays close to $6.0e - 5$ until $n \approx 3.6e + 3$, then reaches a minimum near $1.8e - 5$ before the last update; the run uses about $5.0e + 3$ steps in total. For $s = 0.8$, the plateau value rises to about $7.8e - 5$, with a minimum near $2.3e - 5$; the run uses about $3.5e + 3$ steps. For $s = 0.99$, the plateau value reaches about $9.7e - 5$, with a minimum near $3.0e - 5$; the run uses about $3.0e + 3$ steps. The late reduction of Δt aligns with the singular character of extinction: the curvature $1/R$ rises as R tends to zero, and the characteristic time scale shortens. The final increase of Δt reflects termination at a prescribed final time. Choosing s close to 1 allows Δt to approach $\Delta t_{\max}$ and reduces the number of iterations without degrading the accuracy.

The data in Table 1 also confirm the expected accuracy cost trade-off. The run uses 4611 iterations for $s = 0.6$, 3458 iterations for $s = 0.80$, and 2795 iterations for $s = 0.99$. The choice $s = 0.99$ provides a balanced compromise between efficiency and agreement with the analytic radius for this test.

To rigorously quantify the numerical accuracy, we define the discrete L_2 error norm. Let ϕ_{ij}^n denote the numerical solution and $\phi_{\text{exact}}(x_i, y_j, t^n)$ be the exact analytical solution at time t^n . The discrete L_2 error is defined as:

$$\|e\|_2(t^n) = \left(h^2 \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left| \phi_{ij}^n - \phi_{\text{exact}}(x_i, y_j, t^n) \right|^2 \right)^{1/2}. \quad (31)$$

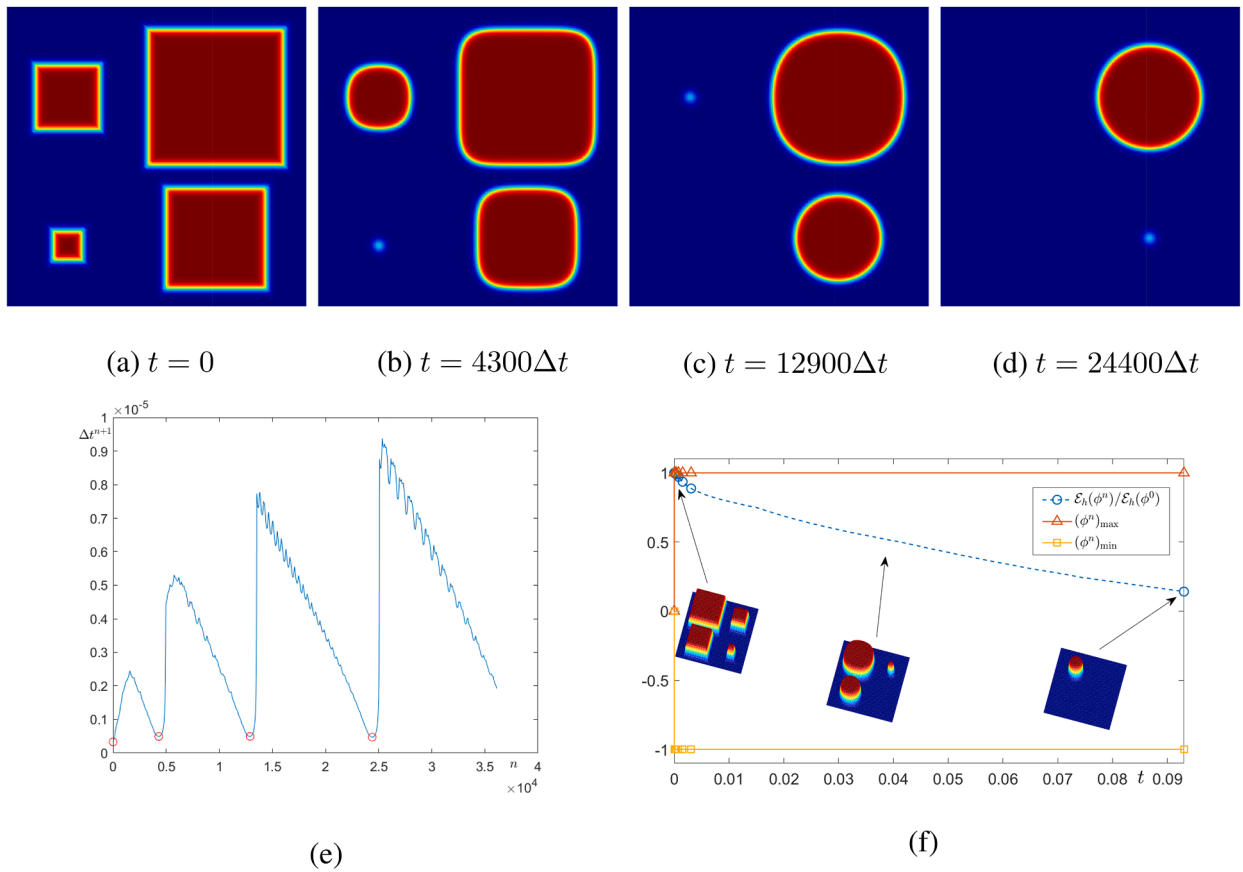


Fig. 4. (a)–(d) snapshots of the temporal evolution of four squares with $\varepsilon = 0.02$ in square domain $(0, 1) \times (0, 1)$ at $t = 0, 4300\Delta t, 12900\Delta t, 24400\Delta t$, respectively. (e) the changes in the time step size Δt^{n+1} . (f) Evolution of discrete energy and maximum principle verification.

Table 1

Sensitivity of the adaptive time-stepping scheme to the safety factor s : total number of accepted iterations in the Fig. 5 experiment.

s	0.6	0.8	0.99
Total steps	4611	3458	2795

For the shrinking circle benchmark, the exact interface motion follows $R(t) = \sqrt{R_0^2 - 2t}$. Given the direct correlation between the phase-field profile and the interface location, we utilize the effective radius error $e_R = |R_{\text{num}} - R_{\text{exact}}|$ as a representative proxy for the global solution accuracy in this specific test.

Fig. 6 illustrates the strategic adjustment of the time step size by the proposed algorithm. The comparison highlights three distinct operational phases. In the initial phase (Iteration steps 0 to ≈ 2200), the solution evolves steadily due to the moderate interface curvature. Consequently, the adaptive algorithm selects a large time step and closely approaches the stability limit $\Delta t_{\max}$. Notably, the adaptive step size (cyan solid line) consistently exceeds that of the aggressive fixed scheme (blue dashed line). This observation confirms that the proposed method utilizes computational resources more effectively than fixed schemes when the solution dynamics permit. As the circle shrinks, the interface curvature increases rapidly, which intensifies the reaction–diffusion dynamics. The adaptive controller detects this physical change via the maximum-norm criterion and enforces a sharp reduction in Δt (visible after iteration 2200). The time step drops significantly below the level of the conservative fixed scheme (red dotted line). This systematic deceleration ensures high temporal resolution precisely when the phase transition dynamics peak. Immediately after the interface extinction, the system reaches a trivial equilibrium state. The algorithm responds to this relaxation by rapidly restoring Δt to the maximum admissible value.

Table 2 summarizes the quantitative performance metrics. The data reveals the dual advantage of the adaptive approach: The proposed scheme requires the fewest time steps (2774), which surpasses even the aggressive fixed scheme (2785 steps). This efficiency stems from the ability to utilize a higher safety factor ($s = 0.99$) in smooth regions, whereas fixed schemes typically require a larger

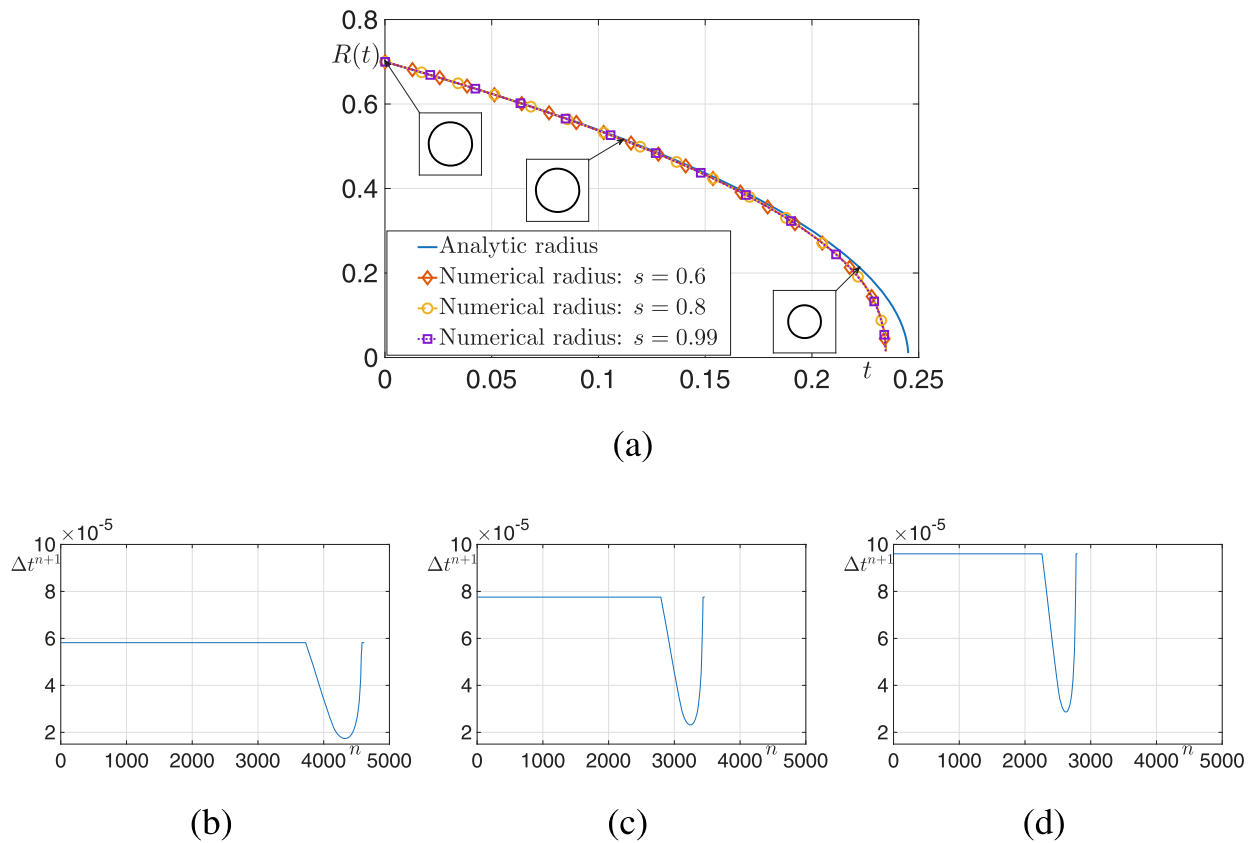


Fig. 5. (a) temporal evolution of a circle. (b)–(d) histories of time step size Δt^{n+1} when $s = 0.6$, $s = 0.8$ and $s = 0.99$, respectively.

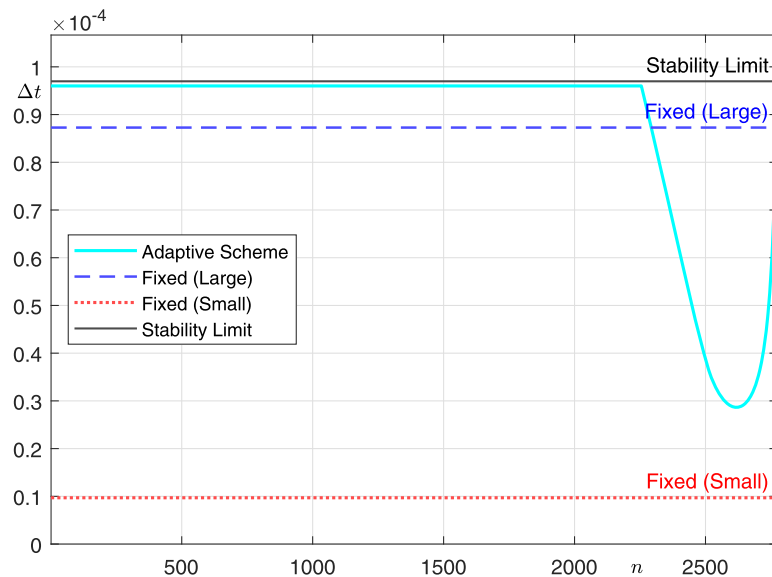


Fig. 6. Temporal evolution of the time step size Δt during the shrinking circle benchmark. The red solid line represents the trajectory of the proposed adaptive scheme. The blue dashed and green dotted lines correspond to the fixed time-stepping schemes with large ($\Delta t = 0.9\Delta t_{\max}$) and small ($\Delta t = 0.1\Delta t_{\max}$) step sizes, respectively. The black solid line indicates the theoretical stability limit $\Delta t_{\max}$.

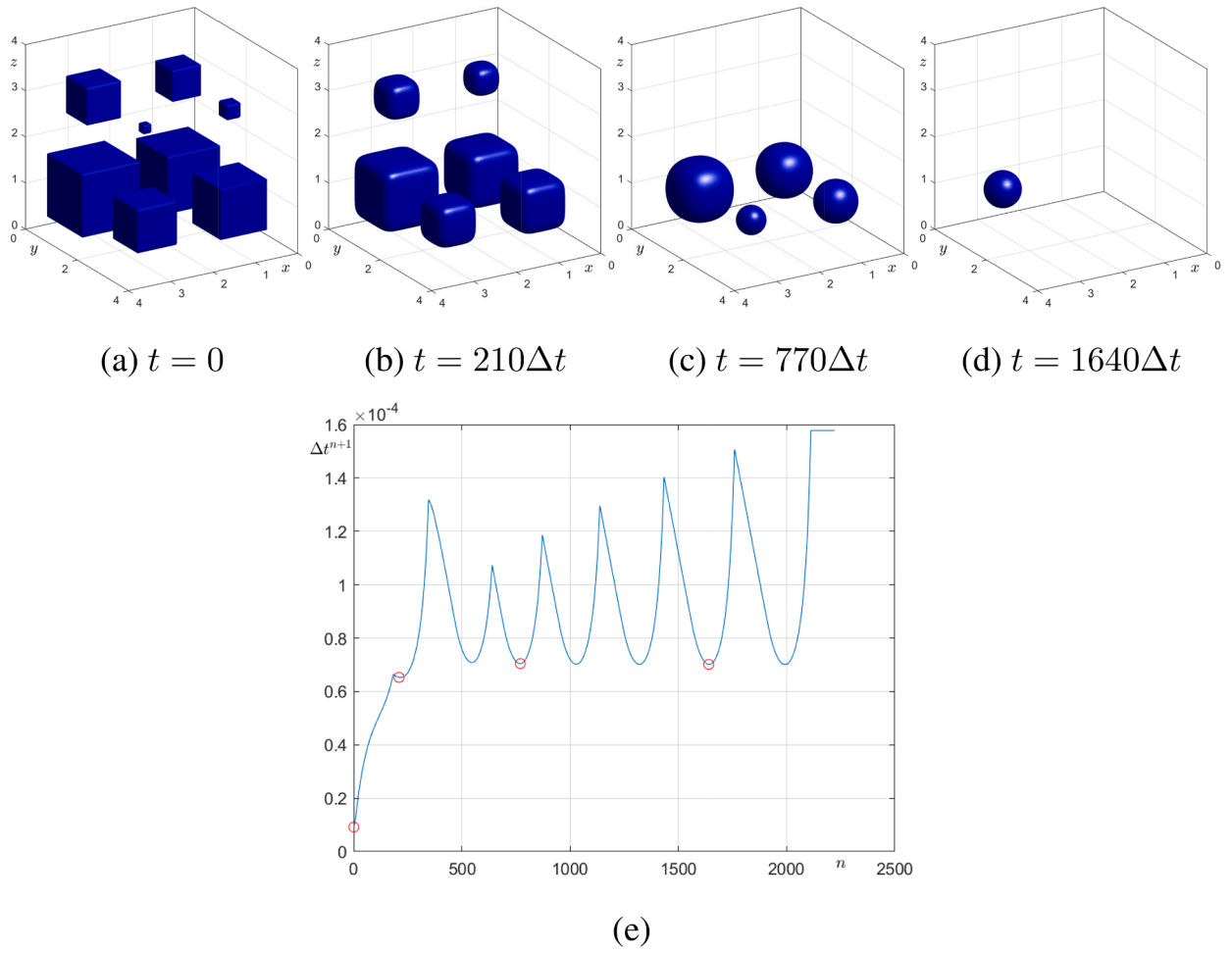


Fig. 7. (a)-(d) snapshots of the temporal evolution of eight regular hexahedrons at $t = 0, 210\Delta t, 770\Delta t, 1640\Delta t$, respectively. (e) the changes in the time step size Δt^{n+1} .

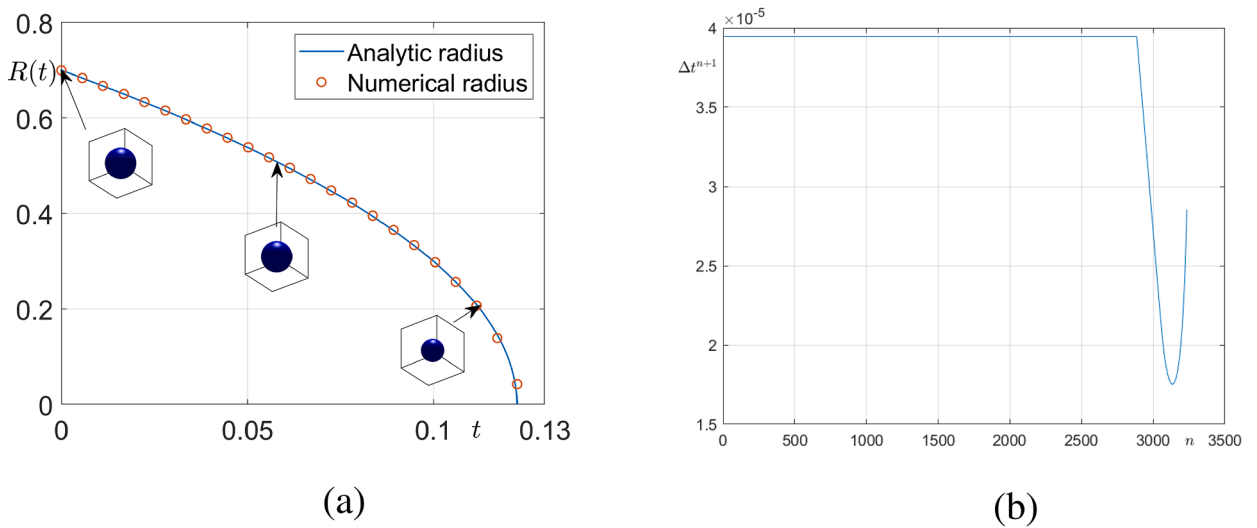


Fig. 8. (a) temporal evolution of a sphere. (b) time step size of numerical radius.

Table 2

Comparison of computational efficiency between the proposed adaptive scheme ($s = 0.99$, $tol = 4.0e - 3$) and fixed time-stepping schemes.

Method	Total steps	CPU time (s)	Radius Error (e_R)
Fixed ($\Delta t = 0.1\Delta t_{\max}$)	25,060	8.9208	1.2346e-02
Fixed ($\Delta t = 0.9\Delta t_{\max}$)	2785	0.9644	1.2105e-02
Proposed ($\Delta t = \Delta t^{n+1}$)	2774	1.0177	1.1972e-02

safety margin to ensure global stability. Consequently, the CPU time of the adaptive method remains comparable to the fastest fixed scheme. Despite the high computational speed, the adaptive scheme yields the lowest L_2 error ($1.1972e - 2$) among all tested methods. While the conservative fixed scheme maintains a constant small step, the adaptive scheme automatically refines the step size to values much smaller than $0.1\Delta t_{\max}$ near the singularity (as shown in Fig. 6). This targeted refinement enables the solver to capture rapid topological changes with superior precision.

3.2. Three-dimensional space

In Fig. 7, we consider the cubic domain $\Omega = (0, 4)^3$ with $128 \times 128 \times 128$ mesh. The parameter values are used as $N_x = 100$, $N_y = 100$, $N_z = 100$, $h = 1/N_x$, $tol = 5.0e - 3$, $\epsilon = h$, $\Delta t_{\max} = \frac{\epsilon^2 h^2}{2h^2 + 6\epsilon^2}$ and the final time T is 0.2. Initial condition is defined as eight regular hexahedrons with sides of different length as shown in Fig. 7(a). The eight regular hexahedrons are rounded and the small two regular hexahedrons disappear in Fig. 7(b). The regular hexahedrons changed into spheres, and the spheres of different sizes were formed in Fig. 7(c). Subsequently, the spheres shrink and vanish in Fig. 7(d). Fig. 7(e) shows the change in time step size Δt^{n+1} . Initially, Δt^{n+1} is set to a small value. As the hexahedrons become more rounded, Δt^{n+1} increases. When the rounded hexahedrons start shrinking, Δt^{n+1} decreases, and as the rounded hexahedrons disappear, Δt^{n+1} increases again.

In Fig. 8(a), analytic radius in three dimensions at time t is given as $R(t) = \sqrt{r^2 - 4t}$ and the initial condition of numerical radius is defined as

$$\phi(x, y, z, 0) = \tanh \left(\frac{r - \sqrt{(x-0.5)^2 + (y-0.5)^2 + (z-0.5)^2}}{\sqrt{2}\epsilon} \right)$$

on $\Omega = (-0.5, 1.5) \times (-0.5, 1.5) \times (-0.5, 1.5)$ with $128 \times 128 \times 128$ mesh, where $r = 0.7$, $h = 2/N_x$, $\epsilon = 4h$, $tol = 5.0e - 3$, $\Delta t_{\max} = \frac{\epsilon^2 h^2}{2h^2 + 6\epsilon^2}$. Fig. 8(a) shows the change of $R(t)$ of analytic radius and numerical radius until $t = 0.1225$. Fig. 8(b) shows time step size of numerical radius during the simulation.

4. Conclusion

In this article, we presented an adaptive time-stepping finite difference algorithm for the AC equation based on an explicit Euler update with a tolerance-controlled step-size selection. The computational experiments demonstrated that the method selects larger time steps during gradual phase changes and smaller time steps during rapid phase changes, which yields improved efficiency over fixed-step explicit schemes. The algorithm has a simple structure, and the manuscript provides Algorithms 1–2 and MATLAB source code.

Limitations and future works. The proposed method is first-order in time and subject to a stability cap $\Delta t_{\max}$, and its adaptivity does not provide an embedded local error estimate or an unconditional discrete energy law. Future work will resolve these limitations through higher-order time integrators, standard error-controlled adaptivity, and energy-stable variants for gradient-flow models. In particular, higher temporal accuracy can follow from replacing the forward Euler step with a higher-order method under the same stability restriction. For example, strong-stability-preserving Runge–Kutta (SSP-RK) schemes [30–32] provide second- or third-order temporal accuracy and preserve nonlinear stability properties under suitable step size constraints. For adaptive accuracy control, step-doubling via extrapolation or embedded RK pairs can furnish a local error indicator that guides step size selection toward a prescribed tolerance. On the spatial side, higher-order finite difference stencils or mesh adaptation near diffuse interfaces can reduce spatial errors at a moderate additional cost. A more detailed convergence analysis of the adaptive mechanism also remains of interest. In particular, a proof that couples the variable-step selection with the partial differential equation limit and quantifies the interplay among tol , $\Delta t_{\max}$, and the global error will strengthen the theoretical foundation.

CRedit authorship contribution statement

Hyundong Kim: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Juho Ma:** Investigation, Software, Validation, Visualization, Writing – original draft; **Hyunho Shin:** Investigation, Software, Validation, Visualization, Writing – original draft; **Junseok Kim:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing.

Conflict of interest

The authors declare that there are no conflicts of interest.

Code Availability Statement

The code used to generate the findings of this study is openly available in GitHub at https://github.com/hdkim4681/Adaptive_time_AC/blob/main/Adaptive_time_AC_MATLAB

Declaration on the use of AI tools

This article was not created with Artificial Intelligence (AI) tools.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.

The MATLAB source code used to generate Fig. 3 is provided.

```

clear all; close all; rng(23)
%% Parameters setting
xl=0; xr=1; yl=0; yr=1; Nx=100; Ny=Nx; h=(xr-xl)/Nx;
x=linspace(xl-0.5*h,xr+0.5*h,Nx+2);
y=linspace(yl-0.5*h,yr+0.5*h,Ny+2);
ep=h; dt=0.01*h^2; tol=2.0e-2; t=0; T=0.25; s=0.99;
max_dt=ep^2*h^2/(2*h^2+4*ep^2);
%% Initial condition
phi0=0.5*(2*rand(Nx+2,Ny+2)-1);
phi=phi0; nphi=phi;
figure(333);
surf(x,y,phi); view(2); shading interp; axis image; axis off; set(gca, ...
    'Position', [0 0 1 1]);
I=2:Nx+1; J=2:Ny+1;
it=0; plotb=30; plotc=6240; plotd=15000;
%% Main loop
while t<T
    it=it+1;
    mu=(phi(I,J).^3-phi(I,J))./ep^2-(phi(I-1,J)+phi(I,J-1)...
        -4*phi(I,J)+phi(I+1,J)+phi(I,J+1))/h^2;
    en=max(abs(mu),[],'all');
    dt=s*min(tol/en,max_dt);
    tt(it)=dt;
    if (t+dt>T)
        dt=T-t;
    end
    t=t+dt
    phi(I,J)=phi(I,J)-dt*mu;

    phi(1,:)=phi(2,:); phi(Nx+2,:)=phi(Nx+1,:);
    phi(:,1)=phi(:,2); phi(:,Ny+2)=phi(:,Ny+1);
    % plot1
    if it==1||it==plotb||it==plotc||it==plotd
        figure(it); clf; hold on
        surf(x,y,phi); view(2); shading interp; axis image; axis off; ...
            set(gca, 'Position', [0 0 1 1]);
    end
end
% plot2
figure(2); clf; hold on; box on;
plot(tt,'-');
plot(1,tt(1),Marker="o",Color=[1 0 0]);
plot(plotb,tt(plotb),Marker="o",Color=[1 0 0]);
plot(plotc,tt(plotc),Marker="o",Color=[1 0 0]);
plot(plotd,tt(plotd),Marker="o",Color=[1 0 0]);
text('Interpreter','latex','String','$\Delta ...
    t^{n+1}$','Position',[-1500,1.7e-05]);
text('Interpreter','latex','String','$n$','Position',[14860,-0.75e-6]);

```


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